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CUP AND BLANK FOR FORMING

Abstract

A blank (10) for forming a cup comprises opposing first and second surfaces (12, 14); opposing first and second side edges (16, 20); opposing top and bottom edges (24, 26); a rim portion (28) extending across the top edge; and an edge tape (30) extending along the first side edge (16) from the bottom edge (26) toward the top edge (24), with the edge tape (30) folded over such that the edge tape extends along the first surface (12) and second surface (14). The first side edge (16) extends such that the edge tape (30) folds over onto itself in at least part of the rim portion.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates to a cup to be sealed on an upper surface of the rim, a blank for forming such a cup and related methods of making.

BACKGROUND OF THE INVENTION

[0002] Any discussion of the prior art throughout the specification should in no way be considered as an admission that such prior art is widely known or forms part of the common general knowledge in the field.

[0003] Cups or containers for food or drink, such as rice, noodles, beans, lentils or other foods made of paper materials are typically formed from a blank, a flat sheet of paper or cardboard which has been cut to the proper dimensions, such as the blank **2** shown in FIG. **1**. One or both of the major surfaces are typically coated, for example with a polymer layer such as Polyethylene, to ensure that the cup will stand up to liquids. For manufacture, the blank is usually rolled up to form a conical or cylindrical shape and the side edges are heat-sealed together (though adhesive or another method could be used). An edge tape **4** can be applied to the side edge which will be interior after the rolling and forming of the cup to help seal at that point. An upper part **6** of the blank is then rolled down to form the upper rim, and a base is inserted and also sealed into place to form the cup. The edge tape **4** extends from the bottom edge to the point at which the rim portion starts. The side edges are also not typically sealed in the rim area so that it is easier to form the rim portion.

[0004] After formation of the cup, the contents can be added, and the top can be sealed. This is often done with a plastic coated aluminum top, which is heat sealed to an upper surface of the rim. The polymer layer on the blank bonds to the plastic layer of the top, sealing the top to the cup.

[0005] JP6060561B2 relates to a cup-shaped paper container in which a flange portion for sealing a lid member has a reduced level difference.

SUMMARY OF THE INVENTION

[0006] In one aspect of the present invention, a blank for forming a cup is provided. The blank comprises opposing first and second surfaces; opposing first and second side edges; opposing top and bottom edges; a rim portion extending across the top edge; and an edge tape extending along the first side edge from the bottom edge toward the top edge. The edge tape is folded over such that the edge tape extends along the first surface and second surface. The first side edge of the blank extends such that the edge tape folds over onto itself in at least part of the rim portion.

[0007] Such a blank can provide a simple way to form a cup that will provide a stable and effective seal with a lid that is heat sealed to a rim of the cup. The part of the edge tape which folds over onto itself will form a part of the rim of the cup, and the extra edge tape will provide extra material for the seal to the lid, particularly at a part of the rim that has overlap between the side edges and a ledge from the side edge.

[0008] According to an embodiment, the first side edge extends in a straight line from the bottom edge to the start of the rim portion and then extends at an angle from the straight line portion. The angle can be angled inward with respect to the straight line part of the side edge, providing the space for the edge tape to fold over onto itself. The angle could be in the range of 10 degrees to 60 degrees, preferably 15 degrees to 45 degrees from the straight line portion of the side edge, though could also vary from these ranges. In some embodiments, the second side edge could be angled in the rim portion in a similar manner.

[0009] According to an embodiment, the edge tape extends to the top edge of the blank. In other embodiments, the edge tape ends before the top edge of the blank, for example at 50% of the height of the rim portion of the blank. The edge tape should extend up to a position where it will be present in a free-standing form (folded over on only itself) at an upper portion of the rim when the cup is formed. Thus, the extension of the edge tape can depend on various factors, such as the thickness of the blank, the dimensions of rim, etc.

[0010] According to an embodiment, at least one of the first and second surfaces comprises a coating. This can be, for example, a waterproof polymer coating such as a polyethylene extrusion coating. Coating at least one (e.g., the inside surface), or both major surfaces helps the cup stand up to any liquid or other materials which may damage or degrade the cup before or during use.

[0011] According to an embodiment, the edge tape is a polymer (e.g., polyethylene) and/or laminate structure, for example, PE/Polyethylene terephthalate (“PET”)/PE or PE/Ethylene vinyl alcohol/PE. Such a material can provide good sealing properties over the interior edge of the blank when forming the cup, thereby helping to preserve the strength and stability of the cup before and during use. Such materials also provides a sturdy free-edge of the edge tape when folded over onto only itself avoiding any problems with folding over or not laying correctly for cup formation, and also delivers a strong seal after heat-sealing to the lid.

[0012] According to a further aspect, a cup is formed with the blank of any of the preceding claims. The blank is rolled into a circular or conical sleeve such that the first and second side edges overlap with the first side edge with edge tape in an inside position. The blank is sealed vertically along at least part of the overlapping portion, for example, the bottom edge to the start of the rim portion. The rim portion is rolled or folded over to form a cup rim. A bottom portion is connected at a base of the sleeve to form the cup. Optionally, the cup further comprises a lid heat-sealed to the rim, for example, a lid with an outer polymer layer for heat-sealing to a polymer layer on the surface of the blank forming the cup, and particularly the upper part of the rim.

[0013] Such a cup is simple to form, and provides a configuration that will be both stable and have a strong seal between the rim and a heat-sealed lid due to the edge tape configuration.

[0014] According to an embodiment, the edge tape folded over onto itself forms a part of an upper portion of the cup rim. This free-standing edge tape part provides additional sealing material at the location where the rim will have a ledge due to the overlap of the blank parts, thereby ensuring that the cup is able to seal to a lid without any holes or gaps.

[0015] According to a further aspect of the invention, a method comprises obtaining a cup according to any of the previously described aspect; and heat-sealing a lid to the rim of the cup. Optionally, the method further comprises filling the cup with a food or drink product prior to the step of heat-sealing. Such a method provides a container which will protect and prolong the life of the food or drink product in the cup due to the ability to provide a good seal between the cup and the lid with the configuration of the blank and edge tape used to form the cup.

[0016] According to a further aspect, a method of producing a blank is provided. The method comprises obtaining a blank with opposing first and second surfaces, opposing first and second side edges, opposing top and bottom edges, and a rim portion extending across the top edge; and applying an edge tape along the first side edge such that the edge tape folds over onto the first and second surfaces from the bottom edge toward the top edge and the edge tape folds over onto itself in at least a part of the rim portion. The edge tape can be applied by first heating the edge tape and placing to fold over and seal to the respective surfaces (and itself). Optionally, the method further comprises forming angled corners along the first and/or second side edges at the rim portion. Such a method provides a blank which can be used to form a cup which will be able to heat-seal to a lid without gaps or holes due to the edge tap configuration providing additional sealing material where it will be needed to form a part of an upper part of the rim when the cup is form.

[0017] The invention will now be further exemplified with the following non-limiting figures and examples.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0018] FIG. 1 shows a schematic plan view of a prior art blank;
[0019] FIG. 2A shows a schematic plan view of a blank according to the present invention;
[0020] FIG. 2B shows a close-up view of a portion of FIG. 2A;
[0021] FIG. 2C shows a cross-sectional view at line C;
[0022] FIG. 2D shows a cross-sectional view at point D;
[0023] FIG. 3 shows a diagram of a process for forming a cup;
[0024] FIG. 4A shows a perspective view of a cup formed with the blank of FIG. 4A;
[0025] FIG. 4B shows a close-up view of a portion of the rim portion of the blank forming the cup, prior to forming the rim of the cup;
[0026] FIG. 4C shows a close-up view of a portion of the rim of the cup of FIG. 4A;
[0027] FIG. 4D shows a cross-sectional view of the cup of FIG. 4A with a lid heat sealed to the upper surface of the rim;
[0028] FIG. 4E shows a cross-sectional view of the rim;
[0029] FIG. 5A shows a second embodiment of a blank for forming a; and
[0030] FIG. 5B shows a third embodiment of a blank for forming a cup.

DETAILED DESCRIPTION OF THE FIGURES

[0031] FIG. 1 shows a schematic plan view of a prior art blank 2 with edge tape 4 as previously described.

[0032] FIG. 2A shows a schematic plan view of a blank 10 according to an embodiment of the present invention; FIG. 2B shows a close-up view of a portion of FIG. 2A; FIG. 2C shows a cross-sectional view at line C; and FIG. 2D shows a cross-sectional view at point D.

[0033] Blank 10 includes first and second surfaces 12, 14; first side edge 16 with angled portion 18, second side edge 20 with angled portion 22, top edge 24, bottom edge 26, rim portion 28 and edge tape 30. Blank 10 is typically formed from a central carton or cardboard layer 26 coated on one or both sides with a protective layer 28 (e.g., a water-proof coating such as a polyethylene extrusion coating), as can be seen in FIG. 2C. In some embodiments, the blank can be another paper or fiber material and/or could be formed of recycled materials.

[0034] Blank 10 is cut and scored to be the appropriate size and dimensions to form a container or cup (see FIG. 4A). In the embodiment shown, that includes arc shaped top and bottom edges 24, 26 and inwardly angled cuts 18, 22 in the rim portion 28 of each of the first and second side edges 16 20. The angle of the cuts can vary, but would typically be in the range of 10 degrees to 60 degrees from the straight line portion of the side edges.

[0035] FIG. 2A shows some example dimensions of blank 10. In the example shown, blank 10 has a height H of about 129 mm, rim section has a height H.sub.RS of about 9 mm. Bottom edge 26 has a length L.sub.BE of about 181 mm, and top edge has a length L.sub.TE of about 283 mm between the upper end of each of the angled sections 18, 22. Such a blank can be used to form a cup with a base diameter of about 61 mm and a rim diameter of about 90 mm at an inner rim diameter and about 97 mm at a rim outer diameter. These dimensions are given as examples only, and various configurations and adjustments could be made depending on the blank, cup and expected usage.

[0036] Rim portion 28 is the portion of blank 10 which will eventually be used to form the rim of the cup. Rim portion 28 typically extends 5-15 mm or about 3%-15% of the height of blank 10 along the top edge 24, though could vary depending on cup and rim requirements.

[0037] Edge tape 30 is typically a polymer material (e.g. Polyethylene ("PE") or laminate structures such as PE/Polyethylene terephthalate ("PET")/PE or PE/Ethylene vinyl alcohol/PE) which is folded over the first side edge 16 to extend on each of the first surface 12 and the second

surface **14**. Edge tape **30** typically extends from the bottom edge **26** of blank **10** toward the top edge **24**. In the embodiment shown, edge tape **30** extends beyond the major straight part **17** of the first edge **16**, and about 50% of the height of the angled portion **18**, though in some embodiments could extend different amounts, for example, to the top edge **24** (see FIG. 5A). Because edge tape **30** extends straight, at the angled portion **18** of first side edge **16**, edge tape **30** folds over onto itself at a free portion **31** of the edge tape **30**, as shown in FIGS. 2B and 2D. Edge tape **30** typically extends about 2-4 mm, preferably 3-3.7 mm, more preferably 3.3-3.7 mm on each surface **12**, **14**, though this could vary depending on the thickness of the blank, sealing needs, etc. This specific placement and extension of edge tape **30** results in a cup which can provide a better seal with a lid, as will be described below.

[0038] FIG. 3 shows a process for forming a cup **40** using the blank **10** of FIG. 2A. FIG. 4A shows a perspective view of cup **40**; FIG. 4B shows a close-up view of a portion of the rim portion **28** of the blank **10** prior to rolling down to form the rim of cup **40**; FIG. 4C shows a close-up view of a portion of the rim of cup **40**; and FIG. 4D shows a cross-sectional view of cup **40** with a lid.

[0039] As shown in FIG. 3, the process of forming cup **40** includes starting with blank **10**, which has been coated (on either the inside surface or on each of first and second surfaces **12**, **14**), cut and had edge tape **30** applied as shown in FIG. 2A. In step **32**, blank **10** is rolled into a circular (or conical) formation to form a sleeve, and then heat sealed vertically along the overlap between the first and second side edges **16**, **20** from the bottom edge **26** to the start of the rim portion **28**. First side edge **16** with edge tape **30** is rolled to be on the inside of the cup **40**. Heating can also be done prior to rolling to prepare for sealing.

[0040] Next, in step **34**, a bottom section (blank) can be connected to a bottom of the sleeve to form cup **40**. This can be through adhesive, heat sealing, or any other suitable method.

[0041] Step **36** then involves rolling down the rim portion **28** to form a rim **42** of cup **40**. Rim portion **28** is rolled outward, typically in multiple steps to roll over the rim portion **28** one or more times, and then optionally form a flat rim **42**, as shown in FIG. 4E. At this point, an upper surface of the rim **42** is formed by the rim portion **28** of the first surface **12** of blank **10** rolled outward and by the free-standing portion **31** of edge tape **30**. Part of the free-standing portion **31** of edge tape **30** forms a portion of the upper surface of the rim **42**, specifically at the point where the first side edge **16** (the angled portion **18**) connects over the second side edge **20**.

[0042] In step **36**, a lid **44** can then be connected to the rim **42** by heat sealing to the upper surface of the rim **42**, including the portion formed by the free-standing part **31** of edge tape **30**. The lid **44** is typically a plastic coated aluminium, and the heat sealing seals the plastic coating from the lid **44** to the polymer coating (on the first surface **12**) on the rim **42** and to the exposed part **31** of edge tape **30** at the connection to form a strong seal.

[0043] In prior art cups, such as those made with the blank shown in FIG. 1, the cup was formed in the same manner. Because the edge tape extended only to the start of the rim portion of the blank, there was no edge tape in the formed rim. The overlapping section of the rim (where the first side edge and second side edge overlap to form the cup and rim) was thicker and had a ledge at the edge of the first side edge. This resulted in more difficult and less successful sealing to the lid due to the gap caused by the overlapped edges.

[0044] In the cup shown in FIGS. 4A-4E, by using the blank of FIGS. 2A-D which extends the edge tape **30** into the rim area **28** to form a free-standing portion **31** of the edge tape **30** (where the edge tape **30** folds over onto itself and not the surfaces **12**, **14** (as shown in FIGS. 4B-4C), edge tape **30** provides extra material to form a stronger and more stable seal between the rim **42** and the lid **44**. This extra polymer material at the location where the first side edge **16** and second side edge **20** join together to form the cup **40** provides extra sealing material which will melt to bond to the polymer from lid **44** upon heat-sealing, thereby ensuring that the lid **44** seals all the way around the rim **42**, filling in any gaps or holes where the side edges connect. Thus, by extending edge tape **30** into the rim portion **28** of blank **10**, and having a part **31** where edge tape **30** is free-standing or

only folding over onto itself (in this embodiment through the angled portion **18** of first side edge **16**) which is located at an upper part of the rim **42**, edge tape **30** provides material to ensure a better seal of lid **44** to rim **42** of cup **40**.

[0045] FIGS. 5A-5B show alternative ways of cutting blank **10** and extending edge tape **30** to achieve similar effects to those described in relation to FIGS. 2A-4E. In FIG. 5A, edge tape **30** is extended to a top edge **24** of blank **10**. This provides additional sealing material and can also be an easier blank to manufacture, simply cutting the edge tape to align with the top edge **24** and bottom edge **26** of the blank.

[0046] FIG. 5B shows an embodiment where instead of an angled cut in the rim portion, an L-shaped cut is made. This also provides the area so that edge tape **30** can be free-standing, or fold over onto itself in the rim portion **28**, thereby helping to promote sealing to a lid at the rim **42** after formation of the cup. While a L-shape cut is shown, many other configurations could be used to allow for the free-standing section of edge tape **30**.

[0047] While the invention has been described with reference to exemplary or preferred embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. For example, while the cup is shown as having a circular cross-section, other versions could have a different cross-section, for example, oval, rectangular, etc. The rim size and configuration could also vary (e.g., have a rounded top). In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular or preferred embodiments or preferred features disclosed, but that the invention will include all embodiments falling within the scope of the appended claims.

Claims

1. A blank for forming a cup, the blank comprising: opposing first and second surfaces; opposing first and second side edges; opposing top and bottom edges; a rim portion extending across the top edge; and an edge tape extending along the first side edge from the bottom edge toward the top edge, the edge tape folded over such that the edge tape extends along the first surface and second surface; wherein the first side edge extends such that the edge tape folds over onto itself in at least part of the rim portion; wherein the first side edge extends in a straight line from the bottom edge to the start of the rim portion and then extends at an angle from the straight line portion; and wherein the edge tape extends to the top edge or ends at about 50% of the height of the rim portion.

2.-5. (canceled)

6. The blank of claim 1, wherein at least one of the first and second surfaces comprises a coating.

7. The blank of claim 1, wherein the second side edge extends with a straight line from the bottom edge to the start of the rim portion and then extends at an angle from the straight line portion.

8. The blank of claim 1, wherein the edge tape is a polymer and/or laminate structure.

9. A cup formed with the blank of claim 1, wherein the blank is rolled into a circular or conical sleeve such that the first and second side edges overlap with the first side edge with edge tape in an inside position; wherein the blank is sealed along at least part of the overlapping portion; wherein the rim portion is rolled or folded over to form a cup rim; and further comprising a bottom portion connected at a base of the sleeve.

10. The cup of claim 9, wherein the edge tape folded over onto itself forms a part of an upper portion of the cup rim.

11. The cup of claim 10, and further comprising a lid heat-sealed to the rim.

12. A method comprising: obtaining a cup according to claim 9; and heat-sealing a lid to the rim of the cup.

13. The method of claim 12, and further comprising filling the cup with a food or drink prior to the

step of heat-sealing.

14. A method of producing a blank, the method comprising: obtaining a blank with opposing first and second surfaces, opposing first and second side edges, opposing top and bottom edges, and a rim portion extending across the top edge, wherein the first side edge extends in a straight line from the bottom edge to the start of the rim portion and then extends at an angle from the straight line portion; applying an edge tape along the first side edge such that the edge tape folds over onto the first and second surfaces from the bottom edge toward the top edge; wherein the edge tape folds over onto itself in at least a part of the rim portion, and wherein the edge tape extends to the top edge or wherein the edge tape ends at about 50% of the height of the rim portion.

15. The method of claim 14, and further comprising: forming angled corners along the first and/or second side edges at the rim portion.
